

Deletion of sulfate transporter SUL1 extends yeast replicative lifespan via reduced PKA signaling instead of decreased sulfate uptake

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Abstract

The regulation of cellular metabolism and growth in response to nutrient availability is essential for cell survival and can significantly impact lifespan. Central to the regulation is a class of transporters that sense and transport specific nutrients and transduce the signal downstream to control genes responsible for growth and survival. In this work, we identified SUL1, a plasma membrane transporter responsible for regulating the entry of extracellular sulfate in *S. cerevisiae*, as a key gene for regulating lifespan. We performed a systematic analysis to delineate the downstream mechanism underlying the lifespan extension by SUL1 deletion. Surprisingly, we found that the lifespan extending effect of SUL1 deletion is not caused by decreased sulfate transport. SUL1 deletion mutant exhibited decreased PKA signaling, resulting in a range of downstream effects, including increased stress-protective trehalose and glycogen, increased nuclear translocation of MSN2 and elevated expression of general stress response genes, enhanced autophagy, and reduced expression of amino acid biosynthetic and ribosomal genes. We show that the observed increase in lifespan is dependent on MSN2 and autophagy pathways. Our findings exemplify the influence of nutrient signaling, rather than the nutrient itself, on lifespan regulation and further substantiate the pivotal role of the PKA pathway in this process.

eLife assessment

The study by Long et al. presents **valuable** findings on the role of the SUL1 gene in yeast longevity, proposing that lifespan extension can occur through signaling pathways independent of its sulfate transport function, offering new insights into aging mechanisms with potential implications beyond yeast biology. However, the evidence supporting the uncoupling of SUL1's transport and signaling functions is **inadequate**, relying on limited lifespan analysis without measurements for nutrients and nutrient signaling status. This research is of particular interest to the aging research community, although additional experiments are needed to fully substantiate the claims.

Introduction

The metabolism of nutrients is known to have an important regulatory impact on the physiology and growth of organisms (González and Hall, 2017 [↗](#)). Reduced food intake has been shown to extend the lifespan across many eukaryotes, such as yeast, worms, flies, and mice (Carmona and Michan, 2016 [↗](#); Green et al., 2022 [↗](#)). The regulation of longevity has been documented to encompass distinct routes and regulators, such as the integrated stress response (ISR), AMPK signaling pathway, and mTORC1 signaling pathway. These pathways are activated in response to various nutrient restriction stresses including glucose, amino acid, and lipid restriction (Loewith and Hall, 2011 [↗](#); Wullschleger et al., 2006 [↗](#)). Eukaryotic cells have evolved distinct mechanisms for nutrition sensing and transportation in order to exert regulatory control over various cellular targets. The deregulation of nutrient sensing, which is widely acknowledged as a hallmark of aging, can influence lifespan and healthspan in a variety of animal models (Acosta-Rodríguez et al., 2022 [↗](#); Duran-Ortiz et al., 2021 [↗](#); Fan et al., 2021 [↗](#)).

The involvement of several nutrient transporters has lately been implicated in the mediation of aging or age-related illnesses. For example, the inhibition of glucose transporter SGLT-2 has the potential to delay cellular senescence (La Grotta et al., 2022 [↗](#)). The amino acid transporter SLC36A4 can regulate the mTORC1 signaling in lysosomes of retinal pigmented epithelial cells through its regulation of the amino acid pool (Shang et al., 2017 [↗](#)). Apo E, which serves as a constituent of plasma lipoproteins responsible for lipids transportation, plays a role in several neurodegenerative disorders (Martens et al., 2022 [↗](#)). In addition to organic nutrients, organisms are subject to the regulation of vast amount of inorganic nutrients including phosphate and sulfate. However, the impact of these nutrients transporters on longevity remains largely unexplored.

The yeast *S. cerevisiae* has been widely used as a model organism in the study of molecular mechanisms underlying aging, as well as evolutionarily conserved signal pathways. Yeast cells possess a diverse array of inorganic nutrient sensing systems that facilitate rapid adaptation to fluctuating environment. For instance, the MEP2 system detects ammonium (Van Nuland et al., 2006 [↗](#)), the PHO84 system senses phosphate (Giots et al., 2003 [↗](#)), the FTR1 system is involved in iron sensing, the ZRT1 system detects zinc (Schothorst et al., 2017 [↗](#)), and the SUL1/SUL2 system that senses and transport sulfate (Kankipati et al., 2015 [↗](#)). Moreover, there is substantial evidence from higher eukaryotes indicating the presence of transporters that exhibit an extra role in nutrient-sensing (Hyde et al., 2007 [↗](#); Pérez-Torras et al., 2013 [↗](#); Walch-Liu and Forde, 2008 [↗](#)). Further investigation is required to examine the impact of inorganic nutrients transporters on

lifespan and the underlying mechanisms. Considering the crucial role of sulfur-containing substances in regulating lifespan, such as hydrogen sulfide, and methionine (Choi et al., 2019 [DOI](#)), the sulfate transporters can serve as a good model for such investigation.

Sulfur is widely recognized as a crucial nutrient that exerts significant regulatory influences on cellular metabolism and proliferation (Maw, 1963 [DOI](#)). Sulfur is mostly absorbed and stored within cells in the form of sulfate, facilitated by specialized membrane transporters (Cherest et al., 1997 [DOI](#)). In yeast, the process of enzymatic reduction converts sulfate to sulfide, which is then integrated into organic molecules by the sulfate assimilation pathway (SAP) (Thomas and Surdin-Kerjan, 1997 [DOI](#)). This pathway involves the utilization of two high-affinity transporters SUL1 and SUL2 (Khurana et al., 2000 [DOI](#)). Both SUL1 and SUL2 have been identified as transceptors (Kankipati et al., 2015 [DOI](#)), which can sense sulfur signals and exhibit a robust response to sulfur deprivation by upregulating sulfate transport. Additionally, they play a crucial role as mediators in the activation of the protein kinase A (PKA) pathway in response to nutrient stimuli, thereby governing various cellular processes (Thevelein and de Winde, 1999 [DOI](#)).

In this study, we examined the regulatory mechanism of SUL1 in relation to yeast longevity. We found that SUL1, instead of SUL2 deletion, significantly extends yeast replicative lifespan, and the increase of lifespan resulting from the deletion of SUL1 is not contingent upon sulfate transport. The SUL1 mutant strain displayed several traits indicative of decreased PKA activity, including accumulation of stress-protective carbohydrates, increase of MSN2 nuclear localization and up-regulation of stress response genes, and down-regulation of ribosomal gene expression. We show that lifespan extension depends on the stimulation of MSN2 transcriptional activity and autophagy. Our study provides an example where downstream signaling of nutrient instead of the nutrient itself influences lifespan, which may be general for other nutrient transceptors. Our findings also support PKA pathway's important role for lifespan regulation.

Methods

Yeast strains, plasmids, and media

All *S. cerevisiae* strains used in this study are the BY4741 (*MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0*) background and are listed in Table S1.

Knockout strains were generated by transforming with a PCR product encoding the kanamycin-resistance (KanR) cassette or HisMX6 cassette to replace the ORF of interest. EGFP-labeled strains were generated by inserting an EGFP-HisMX6 cassette amplified from the pYM-N18-GFP-HisMX6 plasmid into the C-terminal of the target gene. All the used primers are provided in Table S2.

The yeast strains were cultivated in YPD complete medium containing 2% glucose, 2% peptone, 1% yeast extract, or synthetic complete medium supplemented with 0.03% essential amino acids and 2% glucose. The glucose restriction media contained the same components as the complete medium except for a reduced concentration of glucose at 0.05%. Yeast strains were grown in selective minimal media at 30°C and shaken at 250~300 rpm.

Replicative lifespan analysis

The strains were pre-cultured overnight on YPD plates. Lifespan analyses were conducted using micromanipulation, following the previously described protocol (Lin et al., 2000 [DOI](#)). All micromanipulation dissections were performed at laboratory temperature.

RNA-seq and differential expression gene analysis

Yeast in the logarithmic growth phase (~ 0.5 OD₆₀₀) was harvested, and RNA-seq experiments were conducted following the previously described protocol on BGISEQ-500 (Beijing Genomics institution). The sequencing data discussed in this publication have been deposited with the NCBI Gene Expression Omnibus and are accessible through GEO Series accession number GSE215287 (<http://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE215287>). The gene expression levels were compared across different samples using Fragments Per Kilobase Million (FPKM) values. The log₂ ratio of the FPKM value was used to determine the fold change in gene expression for each sample. The R software was then used to carry out a differential expression analysis between the samples (Sun et al., 2013).

RNA extraction, reverse transcription and real-time qPCR

The total RNA was extracted using the TRIzol reagent (Invitrogen, #15596026) following the manufacturer's instructions. Subsequently, residual DNA was removed using gDNA wiper mix, and 1 µg of total RNA was reverse-transcribed to cDNA utilizing HiFiScript gDNA Removal RT MasterMix (CWBIO, Jiangsu, China). Quantitative real-time PCR was performed on the 7500 Fast Real-Time PCR System (Applied Biosystems, Foster City, CA, USA) with 2 x SuperFast Universal SYBR Master Mix (CWBIO, Jiangsu, China). Actin was used as the internal control for data analysis. The relative expression of each mRNA was calculated using the $2^{-\Delta\Delta CT}$ method. Each measurement consisted of three technical replicates. All primer sequences were included in Supplementary Table S2.

Transcriptional factors prediction of DEGs

We utilized the regulator function module of the Saccharomyces Genome Database (SDG) to identify transcription factors associated with differentially expressed genes (DEGs) in our transcriptome sequencing data, and ranked their frequency. Transcription factors with higher frequencies were considered more likely to be common regulators of DEGs. The regulatory network of these transcription factors on DEGs was visualized using Cytoscape 3.7.1.

Glycogen and trehalose quantification

A single clone of yeast was inoculated in 5 mL of YPD medium and incubated overnight at 30°C, 200 rpm. After being diluted to a concentration (OD₆₀₀ = 0.1), the culture was incubated for four hours at the same temperature and speed until it entered the exponential growth phase. The cells were rinsed with 1 mL of sterile ice water and the cell concentration was quantified. 1×10^8 cells from the cell suspension were transferred to a 96-well plate, centrifuged and the supernatant was then removed. After being resuspended in 125 µL of Na₂CO₃, the cells were incubated at 95°C for 3 hours while being rotated occasionally. The plates were then cooled to room temperature, the cell suspension was thoroughly mixed before equally divided into two new plates. There were three technical replicates for each strain.

For glycogen measurements, each reaction solution was prepared by combining 188 µL of amyloglucosidase buffer, 62 µL of cell suspension, and 10 µL of freshly prepared *Aspergillus niger* α-amyloglucosidase solution (~ 70 U/mg) in a sodium acetate buffer (pH = 5.2) at a concentration of 0.2 M. The plates were mixed thoroughly and incubated overnight at 57°C.

For trehalose measurements, each reaction solution was prepared by adding 188 µL of trehalase buffer, 62 µL of cell suspension, and 10 µL of porcine trehalase solution (~ 0.007 U) diluted threefold with 0.2 M sodium acetate buffer (pH = 5.2). The plates were then incubated overnight at 37°C.

The quantification of glucose released in the preceding procedures was performed using the Glucose Assay Kit (Sigma-Aldrich, #MAK263) in accordance with the instructions provided by the manufacturer.

Single-cell time-lapse imaging

The cells were cultured overnight in synthetic complete media containing 2% glucose at 30°C until they reached an optical density of approximately 1.0. Subsequently, the cells were diluted ten-fold and further cultivated in a shaker at 30°C for an additional 4 to 6 hours before being loaded into the microfluidic device. The microfluidic device was employed to track single cell, following the previously described protocols (Chen et al., 2020 [DOI](#); Xie et al., 2012 [DOI](#)). The microfluidic chip used for this experiment was generously provided by Prof. Chunxiong Luo from PKU.

Fluorescence imaging

Yeast cells in the logarithmic growth phase was used to observe the nuclear localization of MSN2-EGFP or MSN4-EGFP. The nuclei were stained with DAPI at a final concentration of 2.5 µg/ml and incubated for 30 min at room temperature. Images were captured using Zeiss Axio Observer 7, and image analysis was conducted using the ImageJ software. In brief, the quantification of protein expression abundance in each cell is determined by subtracting the background intensity from the green fluorescence intensity. In each experiment, a minimum of 100 cells were subjected to analysis.

Flow cytometry

The EGFP-labeled strains were incubated overnight, followed by being diluted to an OD₆₀₀ equals to 0.01. The cultures were then grown for an additional duration of 2-4 hours until they attained an OD₆₀₀ of approximately 0.05 in a 96-well plate. Subsequently, flow cytometry was employed to quantify the EGFP signals (FITC channel) in each sample at regular hourly intervals. The cellular concentration of EGFP was assessed by normalizing the EGFP signal with cell size using a forward scattering signal (FSC channel) for individual cells. Mean fluorescence intensity was utilized to compare the differences.

Statistical analysis

The replicative lifespan was analyzed using the Kaplan-Meier method, the survival differences were determined by the log-rank test. Other group differences were evaluated with Student's t-test. The statistical analyses were conducted using SPSS 22.0 and GraphPad Prism 6. Statistical significance was considered at *p* value < 0.05.

Results

SUL1 deletion increases yeast RLS in a sulfate metabolism independent way

In budding yeast, SUL1 and SUL2 serve as two plasma membrane transporters responsible for regulating the import of extracellular sulfate. The SUL1 functions as a transporter with a high affinity for sulfate (Cherest et al., 1997 [DOI](#)). Sulfate is converted into various sulfur-containing substances, including methionine, cysteine, glutathione, and S-adenosylmethionine through the sulfate assimilation pathway (SAP) (Marzluf, 1997 [DOI](#)). SUL2 is the homolog of SUL1 and may function as a redundant sulfate transporter. We first measured the RLS of SUL1 deletion and found that the mutant has a significantly extended lifespan (Fig. 1A [DOI](#)).

We then seek out to understand the mechanism underlying the lifespan extension. Our initial guess was SUL1 deletion decreases the sulfate transport, limiting the availability of sulfur, thus extends lifespan through a mechanism similar to methionine (which also contains sulfur) restriction.

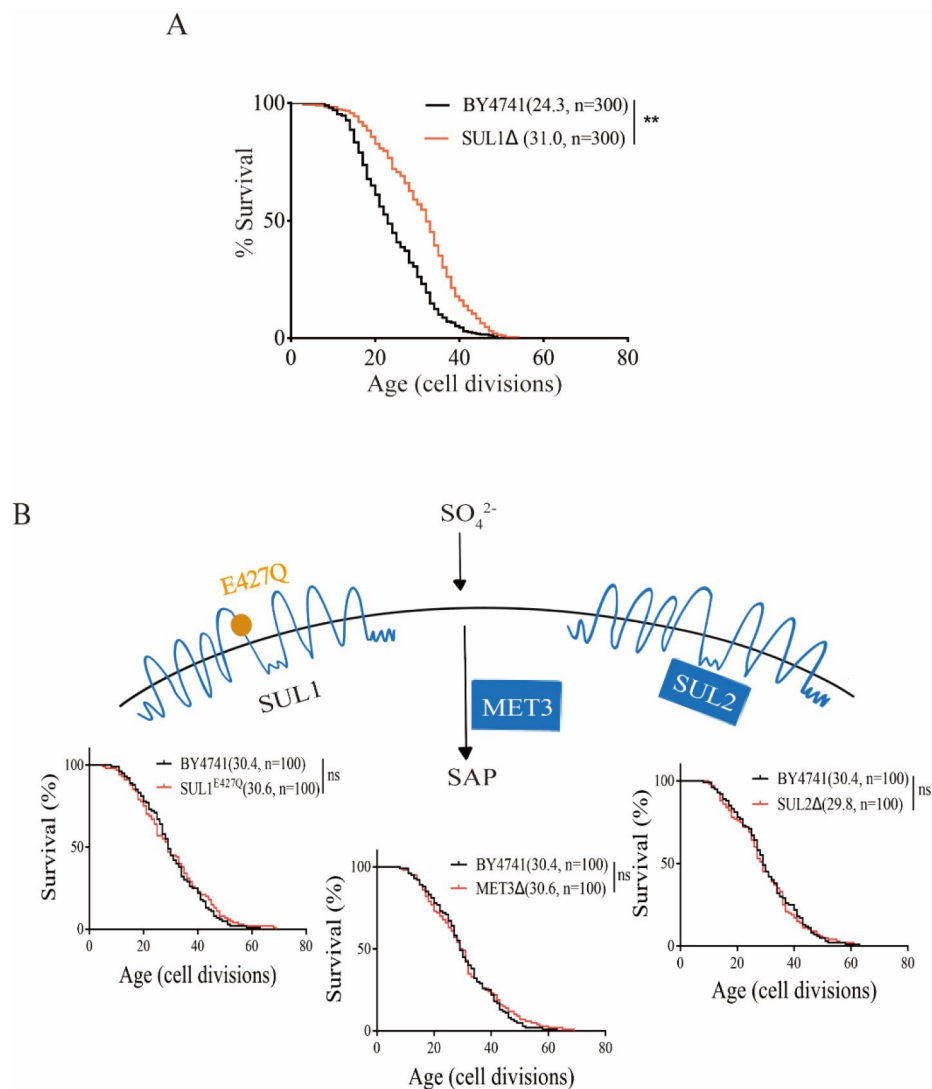


Figure 1.

The lifespan extension of *SUL1Δ* mutant is not caused by changes in sulfate transport/metabolism.

(A) Deletion of *SUL1* gene significantly extended the replicative lifespan of yeast *S. cerevisiae*. Numbers in parentheses indicate the average life span and the number of cells measured. **: $P < 0.01$. (B) Lifespan is not altered by three targeted genetic interventions that change sulfate transport/metabolism: a. mutation of the amino acid residue of *SUL1* (*E427Q*) that abolishes the activity of sulfate transport; b. inactivation of *MET3*, the key enzymes of SAP; c. deletion of *SUL2* (a homolog of *SUL1*). Survival curves for the WT and *SUL1^{E427Q}*, *MET3Δ* or *SUL2Δ* strains are shown.

In order to investigate the influence of sulfur metabolism on the extension of lifespan in the SUL1 knockout strain, we generated three strains that modulate the SAP at various levels. These strains include a SUL2 knockout strain, a site-directed mutant strain SUL1^{E427Q}, which has been previously demonstrated to disrupt sulfate transport (Kankipati et al., 2015), and a MET3 knockout strain that specifically affects a key enzyme in the SAP (Thomas and Surdin-Kerjan, 1997). We found that inhibition of the sulfur-transporting function of SUL1 (SUL1^{E427Q}) does not extend lifespan. In addition, neither the depletion of SUL2 dependent sulfate transport nor the disruption of SAP through MET3 deletion leads to increase in the replicative lifespan of yeast cells (Fig. 1B). These data suggest that lifespan extension by SUL1 deletion is not caused by decreased sulfate uptake.

SUL1 deletion induces global transcriptional changes indicative of conserved longevity promotion mechanisms

In order to gain a deeper understanding of the mechanisms that contribute to the extension of life span in SUL1Δ strains, we conducted a comprehensive analysis of the global gene expression profiles of the SUL1Δ strains in comparison with the wild-type strains through RNA-seq. We observed a wide range of changes in the transcript levels of genes in strains lacking SUL1 (Fig. 2A). A total of 182/57 genes were upregulated/downregulated ($|\log_2 \text{FC}| > 0.5$) (Fig. 2A). Genes up-regulated are significantly enriched for several biological processes (BP), such as transposition, RNA-mediated regulation, ascospore wall assembly, stress response, protein catabolic process, and carbohydrate metabolism (Fig. 2B & 2C left). On the other hand, genes down-regulated are significantly enriched for cellular amino acid biosynthesis, metabolic processes, and ribosome biogenesis (Fig. 2B & 2C right).

Increase in stress response and production of carbohydrates such as trehalose are seen as beneficial factors for promoting longevity (Zhang and Cao, 2017). Additionally, the suppression of protein biogenesis and ribosomal function are also known to extend lifespan (McCormick et al., 2015; Steyfkens et al., 2018). Thus our gene expression analysis suggests that the deletion of the SUL1 gene activates general and conserved anti-aging mechanisms to extend lifespan.

We observed considerable upregulation of several stress response genes, specifically those controlled by MSN2, in the SUL1 deletion mutant. These genes are involved in the synthesis of trehalose and glycogen, such as TPS1, TPS2, GSY2, and others (Fig. 2C, left). To identify potential transcriptional regulators, we performed association analysis between TFs and differentially expressed genes (DEG) (See Methods). We observed that a number of up-regulated DEG in SUL1 mutant are associated with six stress related transcription factors, namely SUA7, MSN2, IXR1, FKH1, FKH2, and SFP1 (Fig. 2D). Interestingly, besides MSN2 which is well known general stress response regulator (Gasch et al., 2000). We also identified IXR1, a transcriptional repressor that regulates hypoxic genes during normoxia, suggesting that SUL1 deletion induces gene expression program similar to diauxic shift, a phenomenon also observed in a constitutively active MSN2 mutant that mimics lower PKA activity (Pfanzagl et al., 2018).

SUL1 deletion inhibits the PKA activity and promotes the nuclear translocation of MSN2

Based on our observations, it is evident that the lifespan extension in the SUL1 mutant does not depend on reduced sulfate transport. It has been reported that SUL1 exhibits the ability to react to environmental stress via signaling through the PKA pathway (Kankipati et al., 2015). Under conditions of nutrient starvation, yeast cells display traits associated with decreased activity of PKA, including the accumulation of stress-protective carbohydrates such as glycogen and trehalose, enhanced resistance to stress, increased expression of stress response gene, and decreased expression of ribosomal genes (Kankipati et al., 2015). Additionally, the transcription factors MSN2/MSN4, which are involved in stress resistance, are suppressed by PKA activity. The

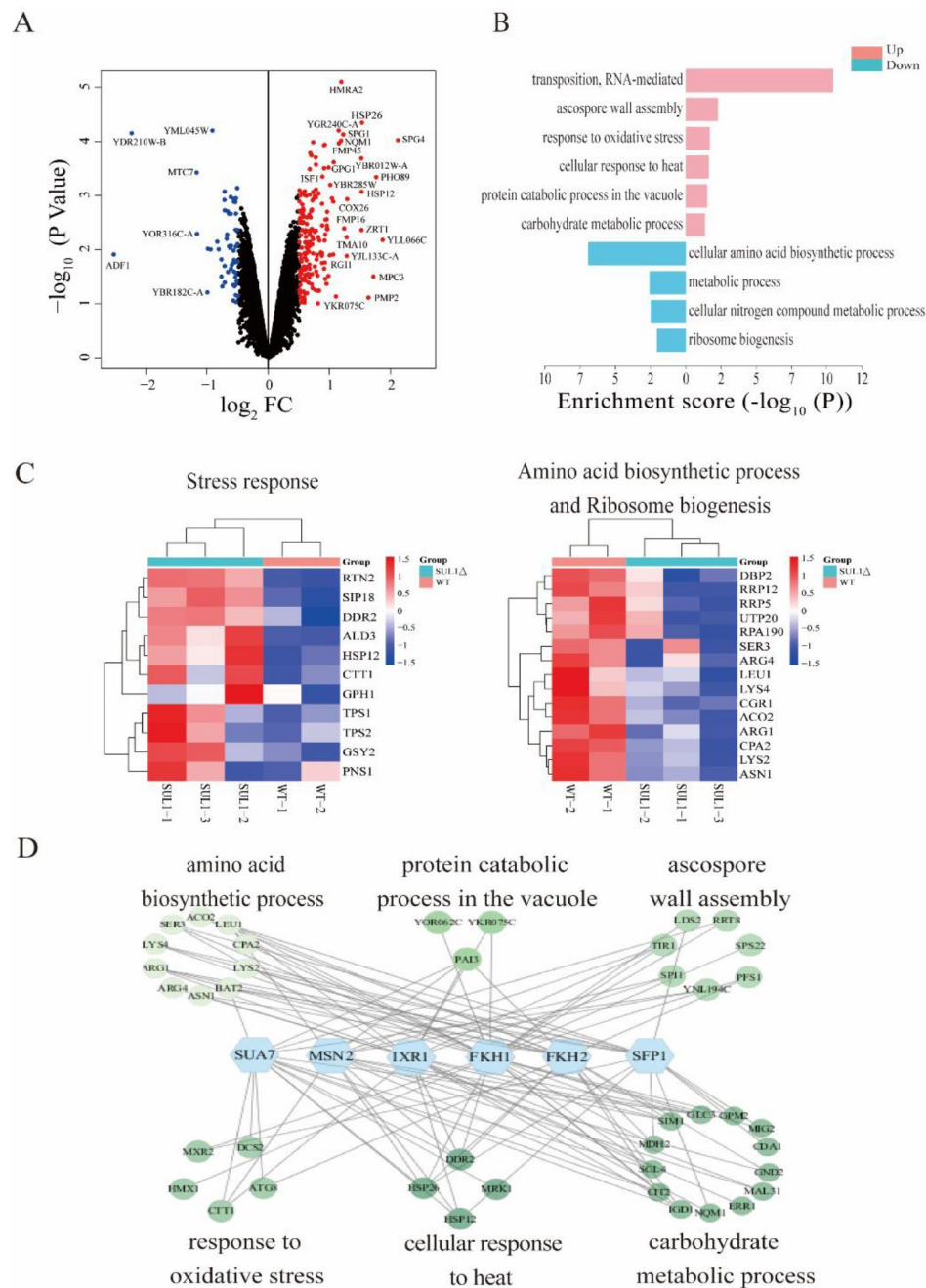


Figure 2.

Common longevity pathways may contribute to the RLS extension of SUL1 deletion mutant.

(A) Volcano plot of the DEG between the SUL1Δ and WT strains. The transcriptomics analysis was performed in the SUL1Δ and WT strains. Log₁₀ of the *P* values is plotted against the Log₂ FC of the FPKM. (B) Biological processes enriched in the DEGs between the SUL1Δ and WT strains. The up-regulated genes ($P < 0.1$, Log₂ FC > 0.5) and the down-regulated genes ($P < 0.1$, Log₂ FC < -0.5) were used for the analysis. (C) Heatmaps showing changes of stress response (left) and amino acid biosynthetic and ribosome biogenesis genes (right). (D) Association analysis of the potential transcription factors and the DEGs in the enriched biological processes (See Methods).

RNA profile data obtained in our study suggests that the deletion of the SUL1 gene is associated with a decrease in PKA activity and an increase in the expression of genes targeted by MSN2. Thus, to understand the downstream mechanisms for the lifespan extension by SUL1 deletion, we turned our attention to the PKA signaling pathway and its influence on the transcriptional control of MSN2.

The down-regulation of the PKA pathway occurs in yeast cells when they are deprived of a crucial nutrient (Thevelein and de Winder, 1999 [\[1\]](#)). This down-regulation leads to adaptive stress protection and the accumulation of trehalose and glycogen (Lillie and Pringle, 1980 [\[2\]](#)). Consequently, trehalose and glycogen serve as a practical indicator for evaluating the activity of the PKA pathway *in vivo* (Schepers et al., 2012 [\[3\]](#)).

We found that the mRNA levels of TPS1 (Trehalose-6-P synthase) and stress-related genes were notably elevated in the SUL1 deletion mutant (**Fig. 3A** [\[4\]](#)). Additionally, the concentrations of trehalose and glycogen were significantly increased, indicating a decrease of PKA activity due to SUL1 deletion (**Fig. 3B** [\[5\]](#)).

We then further investigated the downstream signal of PKA, by examining the expression and the nuclear translocation of MSN2. We constructed an endogenous EGFP-labeled MSN2 in both wild-type and SUL1Δ strains. These strains were then cultured in log phase and subjected to DAPI staining to assess nuclear localization (**Fig. 3C** [\[6\]](#)). The ratio of fluorescence intensity (FI) in the nucleus vs. that in the whole cell was then calculated for each individual cell. We observed that the SUL1Δ strain exhibited significantly increased nuclear accumulation of MSN2-EGFP protein compared to the wild type strain (**Fig. 3D** [\[7\]](#)).

To examine the potential changes in the amount of MSN2 protein during the aging process, we utilized a microfluidic device to dynamically monitor the expression of EGFP-tagged MSN2 in individual cells, in both the SUL1Δ and the wild-type strains (**Fig. 3E** [\[8\]](#)). We did not observe notable changes of the total amount of MSN2 protein in individual cells in the SUL1Δ strain compared to the wild-type strain (Supplementary Fig. S1).

However, we did observe significant increase of nuclear localization of MSN2 over the whole lifespan in the SUL1Δ strain relative to the wild type strain (**Fig. 3F-G** [\[9\]](#)). The SUL1 mutant exhibited a more pronounced change in the nuclear/whole cell MSN2 ratio as the cells age (**Fig. 3F** [\[10\]](#)). It is worth noting that in the early and middle life stages, specifically prior to the 17th generation, there was a notable difference in the nuclear/whole cell MSN2 ratios observed between the two strains. This difference became less significant for the cells passing 17 generations (**Fig. 3G** [\[11\]](#)), possibly due to smaller number of cells that are still alive (25 in the mutant strain and 10 in the wild type strain) (**Fig. 3F** [\[12\]](#)).

In summary, our findings indicate that the deletion of SUL1 results in a decrease in PKA activity and increase in the nuclear localization of MSN2 (moving from cytoplasm to nucleus), consequently up-regulating stress response genes.

Deletion of SUL1 promotes the cellular autophagy

Autophagy is a universal biological process that is required for the lifespan extending effect of various genetic and environmental perturbations (Sampaio-Marques et al., 2014 [\[13\]](#)). In light of the downregulation of the PKA signal in the SUL1 mutant and the upregulation of the vast majority of autophagy genes identified by the transcriptomic analysis (**Fig. 4A** [\[14\]](#)), we investigated the effect of SUL1 deletion on autophagy. We decided to focus on ATG8, initially identified as an essential component of the autophagy machinery (Ichimura et al., 2000 [\[15\]](#)) (Paz et al., 2000 [\[16\]](#)). We generated EGFP-labeled ATG8 in both wild-type and SUL1Δ strains. Using a microfluidic device to monitor the dynamic expression of EGFP-tagged ATG8 in individual cells in the SUL1Δ and the wild-type strains, we found that ATG8 levels steadily increase with age in the SUL1Δ strains, but not in the

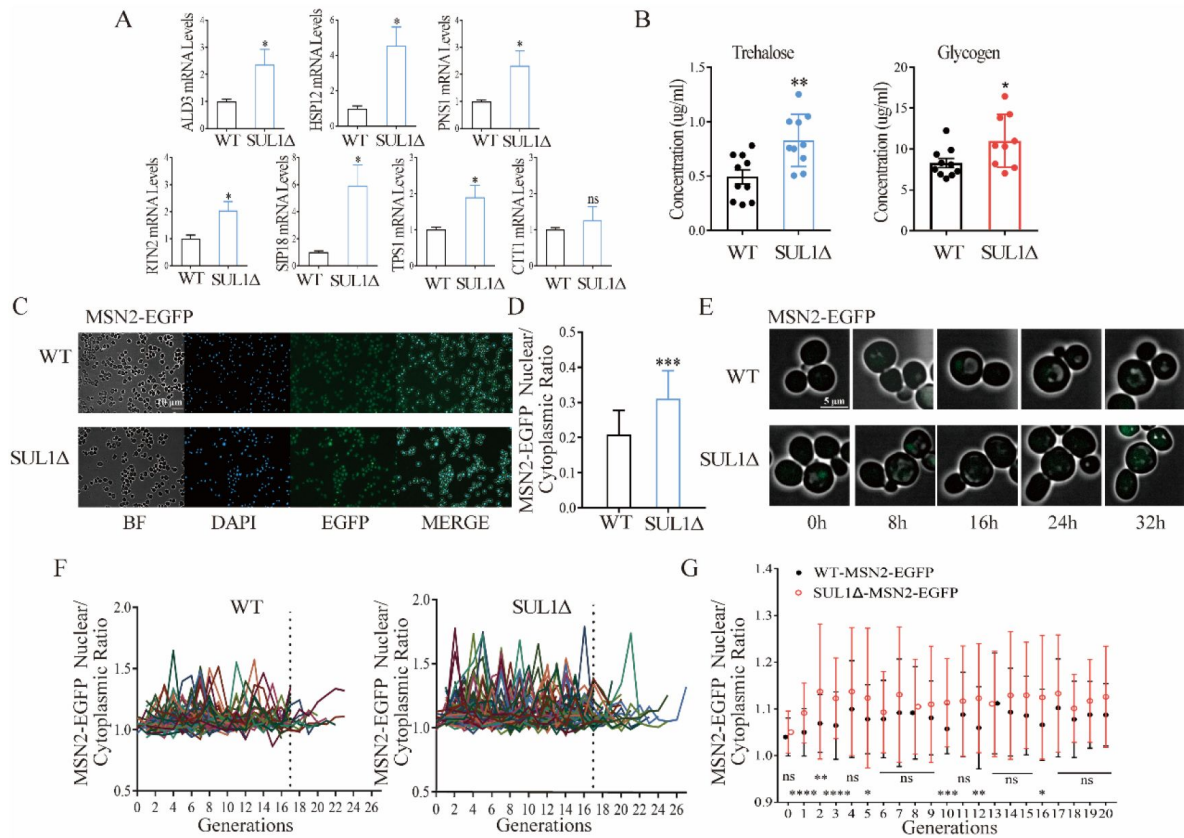


Figure 3.

SUL1 deletion inhibits the PKA pathway and increases the translocation of MSN2 into nucleus.

(A) The mRNA levels of several stress response genes and trehalose synthesis. ns: not significant; *: $P < 0.05$. (B) The concentrations of trehalose and glycogen in WT and SUL1Δ strains. *: $P < 0.05$; **: $P < 0.01$. (C) Representative images of EGFP-labeled endogenous MSN2 in WT and SUL1Δ strains during the exponential growth phase. BF: bright field. Scale bars: 10 μm . (D) The ratio of the mean fluorescence intensity of MSN2-EGFP in the nucleus vs. that of the total cell. Bars represent mean $\pm$ SD, $n = 100$. ***: $P < 0.001$. (E) Representative time-lapse images of MSN2-EGFP in WT and SUL1Δ strains. Scale bars: 5 μm . (F) The normalized nuclear/cytoplasmic fluorescence intensity ratio of MSN2-EGFP as a function of age in the WT and SUL1Δ strains (number of cells WT: $n = 80$; SUL1Δ: $n = 80$). (G) Comparison of the nuclear/cytoplasmic mean fluorescence intensity ratio of MSN2-EGFP as a function of age in WT and SUL1Δ strains. Bars represent mean $\pm$ SD, $n = 80$. ns: not significant; *: $P < 0.05$; **: $P < 0.01$; ***: $P < 0.001$; ****: $P < 0.0001$.

wide type strain (**Fig. 4B** [↗](#) and **4C** [↗](#)). Additionally, a statistically significant difference in ATG8 expression was observed between SUL1Δ and wild-type strains during the middle and late stages of life (**Fig. 4D** [↗](#)).

To further examine the difference in autophagy response between the SUL1Δ and the wild type strains, we introduced glucose restriction as a stress stimulus and dynamically monitored alterations in ATG8 expression before (glucose 2%) and after (glucose 0.05%) the media switch (**Fig. 4E** [↗](#)). Notably, a more pronounced upregulation of ATG8 expression was observed following glucose restriction in the SUL1Δ strain (**Fig. 4F** [↗](#)).

In summary, SUL1Δ strain has elevated expression of autophagy genes and is better at up-regulating autophagy in response to aging and nutrient depletion.

MSN2 and ATG8 are required for the lifespan extension by SUL1 deletion

Given the observation of the elevated nuclear MSN2 level and a stronger ATG8 response during aging in SUL1Δ strain, we further analyzed whether MSN2 and ATG8 are required for the lifespan extension by SUL1 deletion. We compared the RLS of the single mutants MSN2Δ, ATG8Δ, SUL1Δ, and the double mutants SUL1Δ/ MSN2Δ, and SUL1Δ/ ATG8Δ. We found that the deletion of MSN2 or ATG8 alone neither increase nor decrease lifespan. However, deletion of MSN2 (**Fig. 5A** [↗](#)) or ATG8 (**Fig. 5B** [↗](#)) in the SUL1Δ background partially abolish the lifespan extension of SUL1 deletion, arguing that the lifespan extension by Sul1 deletion is at least partially mediated through increased stress response (Msn2) and elevated autophagy (ATG8).

Our findings suggest a mechanistic model in which the PKA pathway plays a pivotal role in regulating lifespan extension in response to SUL1 knockdown. The deletion of SUL1 decreases the activity of the PKA signaling pathway, promoting the accumulation of protective trehalose. By inhibiting ribosome biogenesis, bolstering autophagic capacity, and promoting the translocation of MSN2 and consequently increased stress response, SUL1Δ cells achieve enhanced adaptive stress resistance and increased lifespan (**Fig. 5C** [↗](#)).

Discussion

Nutrient transceptors are a class of transporters that have the dual function of transporting nutrient as well as sensing and transducing the signal. They can regulate intracellular signaling cascades that exert a pervasive influence on cellular processes including autophagy, mRNA and ribosome biogenesis, protein synthesis, glucose metabolism, nucleotide and lipid metabolism, proteasomal activity, and stress tolerance (López-Otín et al., 2023 [↗](#)). Several nutrient-sensing networks have been shown to be associated with longevity in diverse animal models (Acosta-Rodríguez et al., 2022 [↗](#); Duran-Ortiz et al., 2021 [↗](#); Spadaro et al., 2022 [↗](#)). In this study, we provide strong evidence that genetic knockout of SUL1, a sulfate-sensing transceptor, extends the replicative lifespan of *S. cerevisiae* by downregulating PKA signaling to modulate carbohydrate storage, autophagy, ribosome biogenesis, and nuclear translocation of transcription factor MSN2 for general stress response.

Sulfate is transported into the cell of yeast by two high-affinity H-symporters, SUL1 and SUL2 (Breton and Surdin-Kerjan, 1977 [↗](#); Cherest et al., 1997 [↗](#)), where it is reduced to sulfide and homocysteine via the sulfate assimilation pathway. These compounds are subsequently incorporated into sulfur-containing molecules, including methionine, cysteine, glutathione, and S-adenosylmethionine (Marzluf, 1997 [↗](#)). These sulfur-containing molecules function as intracellular methyl donors and can modulate the lifespan of various species (Gu et al., 2017 [↗](#); Obata and Miura, 2015 [↗](#); Walvekar et al., 2018 [↗](#)). Our study revealed, however, that the downregulation of

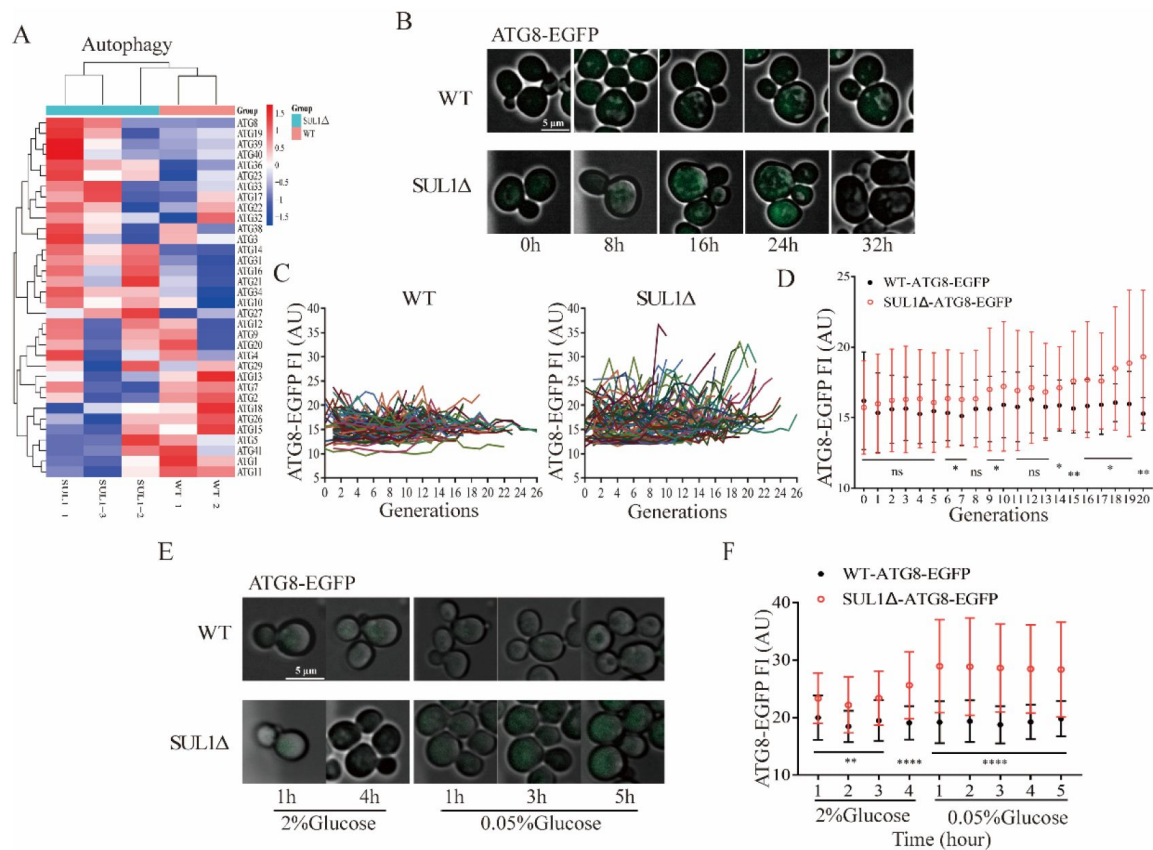


Figure 4.

SUL1 deletion raises cellular autophagy level.

(A) Heatmap showing changes of autophagy-related genes in WT and SUL1Δ strains. (B) Representative time-lapse images of ATG8-EGFP in WT and SUL1Δ strains. Scale bars: 5 μm. (C) The normalized fluorescence intensity of ATG8-EGFP as a function of age in the WT and the SUL1Δ strains. Each colored curve represents a single cell. (number of cells: WT n = 80; SUL1Δ n = 80). (D) Distribution of the fluorescence intensity of ATG8-EGFP as a function of age in the WT and SUL1Δ strains. Bars represent mean ± SD, number of cells n = 80. ns: not significant; *: $P < 0.05$; **: $P < 0.01$. (E) Representative time-lapse images of ATG8-EGFP in WT and SUL1Δ strains grown in complete synthetic medium (2% glucose) or in glucose restriction medium (0.05% glucose) at the indicated times. Scale bars: 5 μm. (F) Distribution of the fluorescence intensity of ATG8-EGFP in WT and SUL1Δ strains grown in complete synthetic medium (2% glucose) or in glucose restriction medium (0.05% glucose) at indicated times. Bars represent mean ± SD; WT: n = 35; SUL1Δ: n = 44. **: $P < 0.01$; ****: $P < 0.0001$.

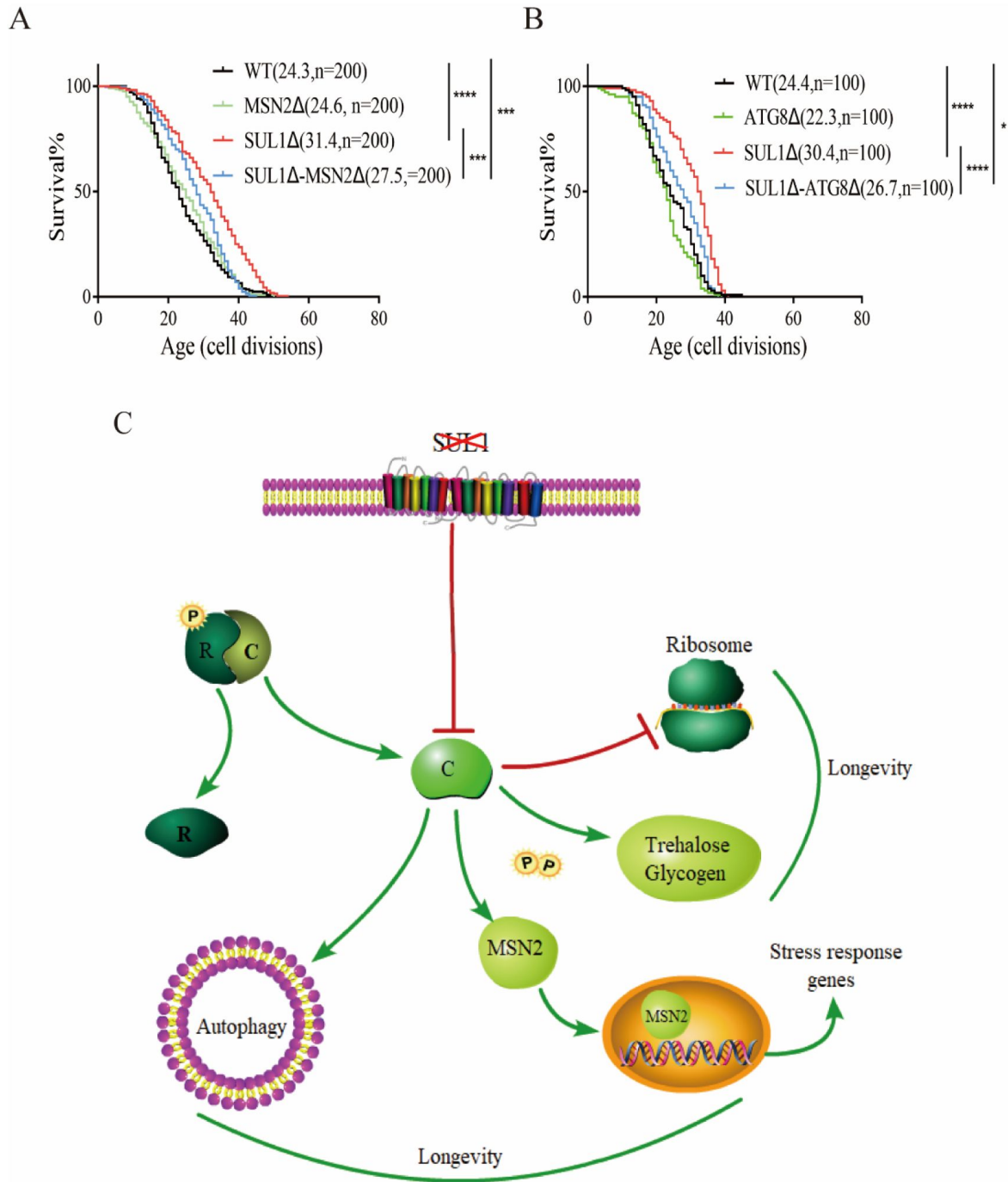


Figure 5.

The effect of SUL1 deletion on longevity is partially mediated by MSN2 and ATG8.

(A-B) Replicative life span of MSN2 and ATG8 deletion mutants in WT and SUL1Δ strains. The median lifespan is displayed on the graph. (C) A schematic illustrating a mechanistic model of how the deletion of SUL1 extends lifespan. SUL1 deletion leads to decreased PKA activity, resulting in increased nuclear translocation of MSN2 (and consequently increased general stress response), autophagy, trehalose, and decreased ribosome biogenesis. The cumulative impact of these downstream effects collectively contributes to the extension of lifespan. R: PKA regulatory subunit; C: PKA catalytic subunit.

the high-affinity transceptor SUL2 did not result in an increase in lifespan. We show that targeted mutation that abolish the sulfate transporter function of SUL1 while preserving its' signaling function did not extend the lifespan. In addition, disrupting the key enzyme involved in the sulfur assimilation pathway did not extend lifespan.

Our study provided an interesting example where lifespan is regulated by the signal transduction downstream of nutrient sensing, instead of the nutrient itself. This might be more general for other nutrient transceptors, and is worth further investigation. There is also an interesting parallel in multi-cellular organism that the lifespan extension by caloric restriction can be abolished by smelling the food (Zhang et al., 2021 [link](#)), highlighting the importance of downstream signaling in regulating lifespan.

Deprivation of a single essential nutrient in a fermentative medium decreases PKA activity (Thevelein and de Winde, 1999 [link](#)), leading to the accumulation of reserve and stress protective carbohydrates, glycogen, and trehalose (Lillie and Pringle, 1980 [link](#)). PKA exerts a pivotal role in nutrient regulation of yeast growth and stress tolerance through its regulation of multiple transcription factors and metabolic enzymes (Thevelein and de Winde, 1999 [link](#)). The functions of differentially expressed genes (DEGs) in SUL1 knockout strains primarily involve RNA-mediated transposition, protein catabolism, carbohydrate metabolism, stress response, cellular amino acid biosynthesis, and ribosome biogenesis, as revealed by our transcriptome analysis. The role of these transposition and metabolic processes in the lifespan extension of SUL1 requires further investigation.

We identified six transcription factors as important regulators of DEGs in SUL1Δ through a computational analysis: SUA7, which is required for transcription initiation (Pinto et al., 1994 [link](#)); IXR1, which regulates hypoxic genes under normoxia conditions (Brown et al., 1993 [link](#)); FKH1 and FKH2, which regulate the CLB2 gene cluster during the G2/M phase of the mitotic cell cycle (Bähler, 2005 [link](#); Jorgensen and Tyers, 2000 [link](#)); SFP1, which regulates ribosomal biogenesis (Blumberg and Silver, 1991 [link](#)); and MSN2, which orchestrates the general stress response (Schmitt and McEntee, 1996 [link](#)).

In consideration of the diminished activity of the PKA signaling, we specifically focused on MSN2 as a downstream transcription factor of PKA and observed a unique dynamic pattern of nuclear localization in the SUL1Δ strain. Regarding the transcriptional activity of MSN2, the SUL1 mutant exhibited a more robust response to both acute and chronic stimulation. The administration of glucose limitation stimulus, which simulates an acute stress in yeast, resulted in a significant enhancement of MSN2 nuclear entry (Supplementary Fig. S2). Compared to the wild type, the translocation of MSN2 into the nucleus was more pronounced in the SUL1Δ strain as cells age, indicating that SUL1 is more sensitive to chronic stresses such as aging-related damage.

In addition to MSN2, MSN4 was initially identified as a PKA controlled transcription factor that regulates stress response and tolerance genes downstream (Görner et al., 1998 [link](#); Martínez-Pastor et al., 1996 [link](#)). MSN2 and MSN4 regulated the transcription of numerous identical downstream targets by binding to the same stress response element, 5'-CCCCT-3' (Mager and De Kruijff, 1995 [link](#)). In this study, we also analyzed in-depth the nuclear translocation of MSN4 to determine its response to stresses. Interestingly, compared to the wild type, no significant MSN4 nuclear translocation alteration was observed in SUL1Δ as a result of aging (Supplementary Fig. S3), indicating that the regulation of stress response capacity of SUL1Δ is MSN2-dependent as opposed to MSN4 dependent.

In addition, PKA signaling regulates the initiation of autophagy pathway (Budovskaya et al., 2004 [link](#)), and ATG8 is recruited to an expanding structure and is required for the completion of autophagosome formation (Xie et al., 2008 [link](#)). Our study revealed that autophagy associated with ATG8 contributed to the lifespan extension of SUL1Δ strain.

In conclusion, our findings demonstrate that downregulation of SUL1 inhibits the activity of nutrient-sensing PKA signaling pathway. This regulatory mechanism is responsible for modulating several biological processes, ultimately leading to an increase in the replicative lifespan of yeast. Since PKA pathway and the associated nutrient response is highly conserved across different organisms (Kankipati et al., 2015 [↗](#)), our study may have significant implications for the regulation of lifespan in more complex organisms. Further research is needed to clarify the regulatory mechanisms that lead to the weakened PKA signaling pathway following SUL1 knockdown, uncover additional biological processes that contribute to the lifespan extension, and investigate the complex interactions between the sulfur signaling pathway and other essential nutrient pathways.

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Author's contributions

J.L., J.Y., H.L. and Y.Z. conceived and designed the experiments. J.L., M.M. and Y.T.C. performed the experiments. J.L. B.G. and J.Y. performed the bioinformatics analysis. J.L., J.Y., H.L. and Y.Z. wrote the manuscript with input from all authors. All authors read and approved the final manuscript.

Competing interests

Authors declare no competing interests in this paper.

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Reviewer #1 (Public Review):

The manuscript by Long et al. focused on SUL1, a gene encoding a sulfate transporter with signaling roles in yeast. The authors claim that the deletion of SUL1, rather than SUL2 (encoding a similar transporter), extended yeast replicative lifespan independent of sulfate transport. They also show that SUL1 loss-of-function mutants display decreased PKA activity, indicated by stress-protective carbohydrate accumulation, relevant transcription factor relocalization (measured during aging in single cells), and changes in gene expression. Finally, they show that loss of SUL1 increases autophagy, which is consistent with the longer lifespan of these cells. Overall, this is an interesting paper, but additional work should strengthen several conclusions, especially for the role of sulfate transport. Specific points include the following:

- What prompted the authors to measure the RLS of sul1 mutants? Prior systematic surveys of RLS in the same strain background (which included the same sul1 deletion strain they used) did not report lifespan extension in sul1 cells (PMID: 26456335).
- Cells carrying a mutant Sul1 (E427Q), which was reported to be disrupted in sulfate transport, did not have a longer lifespan (Figure 1), leading them to conclude that "lifespan extension by SUL1 deletion is not caused by decreased sulfate uptake". They would need to measure sulfate uptake in the mutants they test to draw that conclusion firmly.
- Related to my previous point, another simple experiment would be to repeat the assays in Figure 1 with exogenous sulfur added to see if the lifespan extension is suppressed.
- There needs to be more information in the text or the methods about how they did the enrichment analysis in Figure 2B. P-values are typically insufficient, and adjusted FDR values are reported from standard gene ontology platforms (e.g., PANTHER).

- It is somewhat puzzling that relocalization of Msn2 was not seen in very old cells (past the 17th generation), but it was evident in younger cells. The authors could consider another possibility, that it was early and midlife experiences that made those cells live longer. Past that window, loss of Sul1 may have no impact on longevity. A conditional shutoff system to regulate SUL1 expression would be needed to test the above, albeit this is probably beyond the scope of this report.
- The connections between glucose restriction, autophagy, and sul1 (Figure 4) could be further tested by measuring the RLS of sul1 cells in glucose-restricted cells. If RLS is further extended by glucose restriction, then whatever effects they see should be independent of glucose restriction.
- They made and tested the double (sul1, msn2) mutants, but they should also test the sul1, msn4 combination since Msn4 functions similarly to Msn2.

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Reviewer #2 (Public Review):

Summary:

In this study, the authors find that the deletion of a sulfate transporter in yeast, Sul1, leads to the extension of replicative lifespan. They investigate mechanisms underlying this extension and claim that the effects on longevity can be separated from sulfate transport, and are instead linked to a previously proposed transceptor function of the Sul1 transporter. Through RNA sequencing analysis, the authors find that Sul1 loss triggers activation of several stress response pathways, and conclude that deletion of two pathways, autophagy or Msn2/4, partially prevents lifespan extension in cells lacking Sul1. Overall, while it is well-appreciated that activation of Msn2/4 or autophagy is beneficial for lifespan extension in yeast, the results of this study would add an important new mechanism by which this could be achieved, through perceived sulfate starvation. However, as described below, several of the experiments utilized to support the authors' conclusion are not experimentally sound, and significant additional experimentation is required to support the authors' claims throughout the manuscript.

Strengths:

The major strength of the study is the robust RNA-seq data that identified differentially expressed genes in cells lacking Sul1. This facilitated the authors' focus on two of these pathways, autophagy and the Msn2/4 stress response pathway.

Weaknesses:

Several critical experimental flaws need to be addressed by the authors to more rigorously test their hypothesis.

(1) The lifespan assays throughout the manuscript contain inconsistencies in the mean lifespan of the wild-type strain, BY4741. For example, in Figure 1A, the lifespan of BY4741 is 24.3, and the extended lifespan of the sul1 mutant is 31. However, although all mutants tested in Figure 1B also have lifespans close to 30 cell divisions, the wild-type control is also at 30 divisions in those experiments as well. This is problematic, as it makes it impossible to conclude anything about the lifespan extension of various mutants with inconsistencies in the wild-type lifespan. Additionally, the mutants analyzed in 1B are what the authors use to claim that loss of the transporter does not extend lifespan through sulfate limitation, but

instead through a signaling function. Thus, it remains unclear whether loss of *sul1* extends lifespan at all, and if it does, whether this is separable from cellular sulfate levels.

(2) While the authors use mutants in Figure 1 that should have differential effects on sulfate levels in cells, the authors need to include experiments to measure sulfate levels in their various mutant cells to draw any conclusions about their data.

1. Similar to point 2, the authors focused their RNA sequencing analysis on the deletion of *sul1* and did not include important RNA seq analysis of the specific *Sul1* mutation or other mutants in Figure 1B that do not exhibit lifespan extension. The prediction is that they should not see the activation of stress response pathways in these mutants as they do not see lifespan extension, but this needs to be tested.

(4) While the RNA-seq data is robust in Figure 2 as well as the follow-up quantitative PCR and trehalose/glycogen assays in 2A-B, the follow-up imaging assays for Msn2/4 localization in Figure 2 are not robust and are difficult to interpret. The authors need to include more high-resolution imaging or at least a close-up of the cells in Figure 3C.

(5) The autophagy assays utilized in Figure 4 appear to all be done with a C-terminal GFP-tagged Atg8 protein. As C-terminal GFP is removed from Atg8 prior to conjugation to phosphatidylethanolamine, microscopy assays of this reporter cannot be utilized to report on autophagy activity or flux. Instead, the authors need to utilize N-terminally tagged Atg8, which they can monitor for vacuole uptake as an appropriate readout of autophagy levels. As it stands, the authors cannot draw any conclusions about autophagy activity in their studies.

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Reviewer #3 (Public Review):

Summary:

In this manuscript, Long et al. demonstrated that the deletion of *SUL1*, which encodes a sulfate transporter localized on the plasma membrane, extends the replicative lifespan in *S. cerevisiae*. The authors further investigated the mechanism underlying this lifespan extension. They found that, unlike *sul1Δ* mutants, other mutants that have been shown to have a deficiency in sulfate transport cannot extend lifespan, from which they concluded that it is unlikely that *SUL1* deletion extends lifespan by impairing sulfate intake. The authors then performed a series of characterizations on *sul1Δ* mutants and found that consistent with previous studies, PKA activity is downregulated when *SUL1* is deleted. The authors demonstrated that *SUL1* deletion promotes the nuclear localization of Msn2, as well as autophagy, which are known downstream signals of the PKA pathway. In addition, the authors show that *MSN2* and *ATG8* are indispensable for the lifespan extension in *sul1Δ* cells. Altogether, this manuscript suggests that *SUL1* deletion extends lifespan by affecting PKA activity.

Strengths:

This study reported an interesting phenotype that the deletion of *SUL1*, but not *SUL2*, promotes lifespan extension in budding yeast. The authors performed some characterizations on *sul1Δ* mutants and epistatic studies to demonstrate that this lifespan extension requires *MSN2* and *ATG8*, which further support the importance of the PKA pathway in regulating lifespan.

Weaknesses:

However, one of the major findings in this paper that SUL1 deletion extends lifespan independently of its role in sulfate uptake was merely based on lifespan measurements on *sul2Δ*, *SUL1E427Q*, and *met3Δ* mutants, which cannot exclude the possibility that yeast lifespan is affected by sulfate intake. In addition, the strength of evidence for whether SUL1 deletion extends lifespan through affecting PKA activity is incomplete. It has been shown that Sul1 and Sul2 have redundant functions in both sulfate transport and PKA activation (Kankipati et al. 2015). However, in this manuscript, as shown by the authors, the deletion of SUL2 does not extend the lifespan compared with *sul1Δ* mutants. Without a further characterization on why deletion of SUL1, but not SUL2, extends lifespan, it is likely that SUL1 deletion extends lifespan independently of either sulfate transport or PKA activation.

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Author Response

We appreciate your constructive feedback on our manuscript entitled “Deletion of sulfate transporter SUL1 extends yeast replicative lifespan via reduced PKA signaling instead of decreased sulfate uptake” (ID: eLife-RP-RA-2023-94609). Your comments/suggestions are very helpful for improving our manuscript. In particular, we feel additional experiments and analysis suggested by the reviewers will help strengthen our argument that Sul1 deletion mutant extends lifespan via decreased PKA signaling, instead of via decreased sulfate uptake. Below we outline our response to the reviewer's comments/suggestions and the plans for additional experiments and analysis.

(1) Our current model is that lifespan extension following SUL1 knockout depends on the PKA signaling pathway but not sulfate transport. To further substantiate this, we plan to conduct further transcriptome sequencing and dynamic sulfate uptake experiments using WT, *Sul1D* and *Sul1E427Q* strains. If our model is correct, we expect that PKA signaling pathway will be more repressed in *Sul1D* strain than in *Sul1E427Q* strain, but the sulfate transport will be similar in both strains. This will add strong evidences supporting the model in addition to the lifespan data.

(2) The reviewer mentioned the disparities observed between the lifespan of WT in Figure 1B and other experimental assays. Although it is known that lifespan for WT varies considerably from experiment to experiment (thus the need for WT control for every lifespan measurement), we agree it is important to make a solid conclusion that *Sul1E427Q* does not extend lifespan. We plan to measure the lifespan of more cells for the mutant strains illustrated in Figure 1B and update the data and charts.

(3) Other issues, for example, the small images of Msn2/4 in the nucleus, grammar and formatting errors, and the lifespan data of double (*Sul1/Msn4*) mutants will be addressed in the revised version of the manuscript after we performed the additional experiments/analysis.